

MICHAEL P. RAGSDALE

Systems Architect | Full-Stack Developer | Accessibility Specialist Norfolk, VA (Open to W2 roles within Virginia) hireme@michaelpragsdale.com | michaelpragsdale.com | github.com/raggiesoft

PROFESSIONAL SUMMARY

Systems Architect and Full-Stack Developer driven by a philosophy of "Empathetic Engineering"—the conviction that resilient software must respect both the server's infrastructure and the user's cognitive load. Specializing in bridging the gap between legacy databases and modern, high-performance cloud environments through custom PHP MVC frameworks, Vanilla JavaScript SPA routers, and strict WCAG 2.1 AA compliance. Proven expertise in engineering legally resilient, Section 508 compliant digital infrastructure for enterprise, municipal, and multimedia ecosystems.

TECHNICAL ARSENAL

- **Languages & Scripting:** PHP 8.4, JavaScript (ES6+), Bash / Shell, Python, HTML5, CSS3/SCSS
- **Infrastructure & Cloud:** DigitalOcean (Droplets/Spaces), Nginx, Cloudflare (DNS/SSL), Certbot, Linux (Ubuntu/Debian)
- **Architecture & Automation:** Custom MVC Frameworks, Vanilla JS SPAs, FFmpeg Automation, Decoupled CI/CD Pipelines
- **Compliance & Enterprise:** Strict WCAG 2.1 AA, Section 508, Deque axe, ActiveNet CRM, JSON-LD / Schema.org

RELEVANT TECHNICAL EXPERIENCE

Founder & Systems Architect RaggieSoft Media | Norfolk, VA | Est. 2008 – Present

- **Systems Architecture (Open Source):** Engineered and open-sourced **Stardust Engine CMS** (MIT License), a custom-built PHP 8.4 MVC framework utilizing a file-based JSON architecture to serve massive-scale narrative multimedia while entirely eliminating database latency.
- **Frontend Engineering:** Architected Elara, the CMS's custom Vanilla JS Single Page Application (SPA) router, which parses JSON payloads to dynamically swap entire site themes and UI architectures per route, bypassing traditional monolithic plugin bloat.
- **Deployment Automation:** Designed decoupled Bash CI/CD pipeline agents (Jenna and Sarah) that execute high-frequency, atomic deployments, achieving zero-downtime server synchronization with the frictionless speed of a local file save.
- **Cloud Infrastructure:** Executed a strategic infrastructure migration from restrictive legacy hosting to a high-performance Linux environment on DigitalOcean, secured by Cloudflare edge networking and custom Nginx reverse proxies.
- **Accessibility First:** Embedded strict WCAG 2.1 AA and Section 508 compliance natively into the framework layout, prioritizing "Empathetic Engineering" through dynamic ARIA focus management and cognitive-load reduction.

Technical Lead & Data Normalization Specialist City of Virginia Beach Parks & Recreation | Virginia Beach, VA | May 2023 – Present

- **Legacy Data Normalization:** Engineered a manual ETL (Extract-Transform-Load) pipeline to normalize legacy municipal reservation data from the ActiveNet CRM into fully accessible, digital formats.

- **Compliance Enforcement:** Enforced strict Section 508 compliance for all generated operational documents from a legacy system not designed for such outputs, optimizing internal facility workflows and resource allocation.
- **Technical Training:** Serve as the primary technical trainer for municipal staff on the ActiveNet CRM, establishing strict data integrity protocols and payment processing security.
- **Operational Leadership:** Function as the primary escalation point for resolving complex facility logistical conflicts and overseeing high-volume financial reconciliation, making independent technical and policy-aligned decisions.

Section 508 Compliance Specialist Old Dominion University (ODU Online) | Norfolk, VA | Nov 2019 – Jul 2022

- **Enterprise Accessibility:** Served as the primary accessibility gatekeeper for enterprise-level web content during a massive university-wide migration to the Canvas LMS.
- **Remediation & Auditing:** Enforced strict WCAG 2.1 AA and Section 508 compliance parameters, remediating complex HTML/CSS code and verifying structural integrity using tools like Deque axe.
- **Architectural Standardization:** Designed and engineered a universal, inherently accessible architectural template deployed across university-wide syllabi, significantly streamlining workflows for instructional design faculty.

Mobile Applications Developer Intern Dominion Enterprises | Norfolk, VA | Sep 2012 – Dec 2012

- Engineered cross-platform mobile applications for iOS and Android utilizing the JavaScript-based Titanium SDK.
- Gained foundational, hands-on experience in mobile architecture and deployment, establishing a decade-long trajectory in software engineering.

EDUCATION & CERTIFICATIONS

- **Tidewater Community College** | Virginia Beach, VA Associate of Science, Information Technology (Focus: Software Engineering & Web Dev) | *Expected Dec 2026*
- **Old Dominion University** | Norfolk, VA Bachelor of Science, Customized Studies in Leadership | *In Progress*
- **Wilson Workforce & Rehabilitation Center** | Fishersville, VA Certificate, Business & Information Technology | *2011*